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SECP3623 – DATABASE PROGRAMMING
SEM I – 2025/2026
MongoDB(shopdb.orders)

Instruction:

Insert the following data into the shopdb database and the orders collection.

use shopdb

```
db.orders.insertMany([
  {
    order_id: "ORD101",
    customer: "Ali Ahmad",
    product: "Laptop",
    quantity: 1,
    price: 3500,
    status: "Completed",
    order_date: new Date("2025-01-05")
  },
  {
    order_id: "ORD102",
    customer: "Siti Aminah",
    product: "Smartphone",
    quantity: 2,
    price: 2800,
    status: "Pending",
    order_date: new Date("2025-01-06")
  },
  {
    order_id: "ORD103",
    customer: "John Lee",
    product: "Tablet",
    quantity: 1,
    price: 1800,
    status: "Completed",
    order_date: new Date("2025-01-07")
  },
  {
    order_id: "ORD104",
    customer: "Nur Aisyah",
    product: "Laptop",
    quantity: 1,
    price: 3700,
    status: "Cancelled",
    order_date: new Date("2025-01-08")
  }
])
```

```

},
{
  order_id: "ORD105",
  customer: "Ahmad Zaki",
  product: "Headphones",
  quantity: 3,
  price: 450,
  status: "Completed",
  order_date: new Date("2025-01-09")
},
{
  order_id: "ORD106",
  customer: "Farah Izzati",
  product: "Smartwatch",
  quantity: 2,
  price: 1200,
  status: "Pending",
  order_date: new Date("2025-01-10")
}
])

```

- a) Write a MongoDB command to display all orders with status "Completed".

Output:

```

{
  _id: ObjectId('695fa103553c77fd40a3042c'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042e'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a30430'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
}

```

- b) Write a MongoDB command to display all orders with price greater than RM2000.

Output:

```
_id: ObjectId('695fa103553c77fd40a3042c'),
order_id: 'ORD101',
customer: 'Ali Ahmad',
product: 'Laptop',
quantity: 1,
price: 3500,
status: 'Completed',
order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042d'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042f'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}
```

- c) Write a MongoDB command to display all orders sorted by price in descending order.

Output:

```

    _id: ObjectId('695fa103553c77fd40a3042f'),
    order_id: 'ORD104',
    customer: 'Nur Aisyah',
    product: 'Laptop',
    quantity: 1,
    price: 3700,
    status: 'Cancelled',
    order_date: 2025-01-08T00:00:00.000Z
  }
  {
    _id: ObjectId('695fa103553c77fd40a3042c'),
    order_id: 'ORD101',
    customer: 'Ali Ahmad',
    product: 'Laptop',
    quantity: 1,
    price: 3500,
    status: 'Completed',
    order_date: 2025-01-05T00:00:00.000Z
  }
  {
    _id: ObjectId('695fa103553c77fd40a3042d'),
    order_id: 'ORD102',
    customer: 'Siti Aminah',
    product: 'Smartphone',
    quantity: 2,
    price: 2800,
    status: 'Pending',
    order_date: 2025-01-06T00:00:00.000Z
  }

```

```

    _id: ObjectId('695fa103553c77fd40a3042e'),
    order_id: 'ORD103',
    customer: 'John Lee',
    product: 'Tablet',
    quantity: 1,
    price: 1800,
    status: 'Completed',
    order_date: 2025-01-07T00:00:00.000Z
  }
  {
    _id: ObjectId('695fa103553c77fd40a30431'),
    order_id: 'ORD106',
    customer: 'Farah Izzati',
    product: 'Smartwatch',
    quantity: 2,
    price: 1200,
    status: 'Pending',
    order_date: 2025-01-10T00:00:00.000Z
  }
  {
    _id: ObjectId('695fa103553c77fd40a30430'),
    order_id: 'ORD105',
    customer: 'Ahmad Zaki',
    product: 'Headphones',
    quantity: 3,
    price: 450,
    status: 'Completed',
    order_date: 2025-01-09T00:00:00.000Z
  }

```

- d) Write a MongoDB command to display all "Pending" orders sorted by order_date (latest first).

Output:

```

< {
  _id: ObjectId('695fa103553c77fd40a30431'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695fa103553c77fd40a3042d'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}

```

- e) Using aggregation, calculate the total sales amount (price) for each product.

Output:

```
< {
  _id: 'Smartwatch',
  total_sales_amount: 1200
}
{
  _id: 'Smartphone',
  total_sales_amount: 2800
}
{
  _id: 'Laptop',
  total_sales_amount: 7200
}
{
  _id: 'Tablet',
  total_sales_amount: 1800
}
{
  _id: 'Headphones',
  total_sales_amount: 450
}
```

- f) Using aggregation, calculate the total sales amount for orders with status "Completed".

Output:

```
< {
  _id: 'Total Completed Sales Amount',
  total_sales_amount: 5750
}
```

- g) Write a MongoDB command to update the status of order ORD102 from "Pending" to "Completed".

Output:

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
_id: ObjectId('695fa103553c77fd40a3042d')
order_id: "ORD102"
customer: "Siti Aminah"
product: "Smartphone"
quantity: 2
price: 2800
status: "Completed"
order_date: 2025-01-06T00:00:00.000+00:00
```

- h) Write a MongoDB command to increase the quantity of order ORD105 by 1.

Output:

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
_id: ObjectId('695fa103553c77fd40a30430')
order_id: "ORD105"
customer: "Ahmad Zaki"
product: "Headphones"
quantity: 4
price: 450
status: "Completed"
order_date: 2025-01-09T00:00:00.000+00:00
```

- i) Write a MongoDB command to delete the order with order_id "ORD104".

Output:

```
< {
  acknowledged: true,
  deletedCount: 1
}
```

- j) Write a MongoDB command to delete all orders with status "Cancelled".

Output:

```
< {
  acknowledged: true,
  deletedCount: 0
}
```